Breast Cancer Outcome Prediction Using Machine Learning Models on BCSC Mammography Data

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Abstract

This project evaluates the predictive performance of Logistic Regression and Random Forest models in identifying short-term breast cancer outcomes using screening data from the Breast Cancer Surveillance Consortium. The analysis examines each model's ability to classify whether a cancer diagnosis will occur within one year of screening. Logistic Regression achieved an accuracy of 0.9947 with an AUC of 0.9109, while Random Forest achieved an accuracy of 0.9940 with an AUC of 0.8876. The dataset consists of approximately 20,000 digital and 20,000 film screening examinations collected between 2005 and 2008. A primary limitation is that the dataset is intended exclusively for instructional use in data analysis and epidemiologic study design rather than for primary research (Breast Cancer Surveillance Consortium, n.d.), which restricts the generalizability of the findings.

Keywords: breast cancer, mammography, predictive modeling, machine learning, logistic regression, random forest, classification performance

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Breast cancer remains the leading cause of cancer-related mortality among women worldwide (Key et al., 2001). Although its etiology is not fully understood, a range of screening and risk-assessment methods have been developed to support early detection and to identify individuals whose genetic or clinical profiles indicate elevated risk for future disease.

Mammography, a low-dose X-ray imaging technique, is the standard screening tool used to detect breast cancer at an early stage before symptoms develop (American Cancer Society, 2023). Regular screening has been shown to improve early detection and reduce mortality by enabling treatment to begin earlier in the disease course (National Cancer Institute, 2025). However, a growing body of research shows that mammography alone is not sufficient as a comprehensive early-detection strategy. Although mammography enables earlier identification of many breast cancers, it also increases measured incidence through the detection of cancers that would either have appeared much later or never become clinically relevant. This phenomenon, known as overdiagnosis, results in diagnosis and treatment without survival benefit and imposes substantial physical and ethical burdens on patients (Løberg et al., 2015). Mammography also fails to detect a significant proportion of cancers. Interval cancers, which arise after a normal screening exam and before the next scheduled screen, account for roughly one-third of cancers among screened women and include both overlooked lesions and rapidly developing tumors not visible at the prior exam (Løberg et al., 2015). These limitations demonstrate that mammography performs unevenly across patient groups and cannot function as a standalone method for reliable early detection. Broader screening strategies that incorporate adjunct imaging and individualized risk assessment are necessary to address both the false-negative burden and the risks of overdiagnosis inherent in mammography-only screening.

This project focuses on using Supervised Learning methods to evaluate mammography data and eliminate lapses in human judgement that may cause these misdiagnoses and overdiagnoses. As

mammography became more widely adopted, large observational datasets were developed to evaluate screening performance and inform methodological research. Data for this project is drawn from the Breast Cancer Surveillance Consortium (BCSC) to compare the predictive performance of Logistic Regression and Random Forest models in classifying short-term breast cancer outcomes following screening.

Method

Digital Mammography Dataset

The BCSC's Digital Mammography Dataset is comprised of a random sample of 20,000 digital and 20,000 film-screen mammograms collected between January 2005 and December 2008 and includes variables such as patient demographics, clinical history, imaging characteristics, and radiologist assessments. The dataset contains no imaging files, and all information is derived from radiologist assessments and participant characteristics recorded at the time of screening.

Data Documentation

Each record includes the BI-RADS assessment, a binary indicator of breast cancer diagnosis within one year of screening, and a set of features relevant to mammography performance. These features are comprised of age, family history of breast cancer, breast density, body mass index (BMI), hormone therapy use, biopsy history, prior mammography, availability of comparison films, and mammography modality. Categorical variables follow the BCSC coding scheme, including explicit codes for missing values. Complete coding definitions for all variables are provided in Appendix, Table A1. The modeling outcome is the presence or absence of a cancer diagnosis within one year of the screening exam.

Data Preparation

Data preparation began by converting the dataset's missing-value codes ("9" and "–99") into formal missing entries and assessing the extent and pattern of missingness across all predictors. Records

containing incomplete or invalid values were removed to ensure that both models were trained and evaluated on an identical and internally consistent dataset. Categorical predictors were retained using their documented ordinal or nominal encodings to preserve the structure defined in the original data collection. Two variables, Patient ID (ptid) and Cancer Type (CaTypeO), were removed before fitting the data into the model. Patient ID carried no predictive information and would introduce dependence across repeated screenings for the same individual, while Cancer Type is only defined for diagnosed cases and would leak outcome information into the predictor set. The result was a cleaned dataset that provides a standardized set of clinical and imaging-related variables suitable for evaluating binary classification models.

Supervised Learning Methods

Two supervised learning methods were selected to evaluate short-term breast cancer outcome prediction: Logistic Regression and Random Forest. Logistic Regression provides a parametric, interpretable baseline model that estimates the linear effects on the log-odds a cancer diagnosis, which supports direct examination the contribution of each predictor. On the other hand, Random Forest offers a nonparametric ensemble approach capable of modeling complex interactions and nonlinear relationships that may arise in clinical and imaging-derived variables. Using both methods enables comparison between a transparent, assumption-driven classifier and a flexible, high-variance model that can capture structure that might be missed by a linear specification.

Logistic Regression

The cleaned dataset was split into the target outcome, defined as a binary indicator of whether a breast cancer diagnosis occurred within one year of the screening examination, and a set of predictor variables consisting of all remaining features. Predictors that could introduce dependence or outcome leakage, such as patient ID and cancer type, were removed before fitting the model. An 80/20 train–test

split with a fixed random seed was used to ensure reproducibility and to maintain a consistent evaluation framework across models.

The Logistic Regression model was trained using the standard penalized maximum-likelihood procedure with default regularization and optimization settings. The maximum number of iterations was increased to 1000 to ensure convergence. Predictors remained in their original encodings, allowing the model to estimate linear contributions from both categorical and continuous variables. The fitted model produced both class labels and predicted probabilities for the held-out test set.

Model performance was evaluated using accuracy and the area under the receiver operating characteristic curve (AUC), which together characterize overall classification performance and probability-based discrimination. Coefficients were extracted from the final model and converted into odds ratios to facilitate interpretation of each predictor's direction and magnitude of association with the one-year cancer outcome. This transformation provided an interpretable summary of how individual clinical and screening variables contributed to short-term risk estimation.

Random Forest

The Random Forest model used the same preprocessing steps, outcome definition, predictor set, and 80/20 train–test split applied in the Logistic Regression analysis. This ensures that both models can be compared and evaluated under identical conditions. A Random Forest classifier was trained using 500 decision trees, with the default bootstrapping procedures to generate bootstrap samples for each tree and feature subsampling applied at each split. Trees were allowed to grow without depth restrictions to enable the ensemble to capture complex interactions and nonlinear relationships among predictors.

Each tree was trained on a bootstrap sample of the original training data and generated a classification based on the splits it learned independently. Final model predictions were then produced by combining the outputs of all trees in the ensemble. Class labels were assigned through majority

voting, while predicted probabilities were calculated as the average predicted probability of the positive class across trees.

Model performance was evaluated with the same held-out test set used for the Logistic Regression model, ensuring direct comparability across methods. Accuracy and the AUC were computed to characterize overall classification performance and probability-based discrimination. After model fitting, feature importances were extracted from the ensemble to quantify each predictor's average contribution to impurity reduction across tree splits. These values represent a nonparametric measure of variable influence and reflect how frequently and effectively each predictor improves the homogeneity of nodes within the forest. The resulting importance scores identified which predictors were most relevant to distinguishing between positive and negative one-year cancer outcomes within the ensemble.

Results

Logistic Regression

The Logistic Regression model achieved an accuracy of 0.9947 and an AUC of 0.9109 on the held-out test set. Coefficients displayed consistent directional effects, as shown in Table 1. Biopsy history had the largest positive coefficient (0.58), representing the strongest elevation in one-year cancer odds among all predictors. Family history also showed a meaningful positive association (0.37), indicating increased risk for individuals with a first-degree relative previously diagnosed with breast cancer. These two clinical history variables formed the primary positive contributors to the model's risk estimation.

In contrast, BI-RADS assessment had the most pronounced negative coefficient (-2.27), reflecting a sharp reduction in one-year cancer probability associated with higher assessment categories in this instructional screening dataset. Other predictors, including BMI, age, breast density, prior mammography use, hormone therapy use, modality type, and comparison films, contributed

comparatively modest effects. Their odds ratios clustered close to 1.0, indicating weaker associations with short-term cancer outcomes relative to the dominant clinical and radiologic predictors.

Overall, the odds ratio pattern indicates a model structure driven primarily by biopsy history and family history with demographic and technical screening variables playing smaller roles. Figure 1 presents the corresponding odds ratios, illustrating the relative magnitude of each predictor's contribution.

Random Forest

The Random Forest model achieved an accuracy of 0.9940 and an AUC of 0.8876 on the held-out test set. Feature importances are presented in Table 2 and indicated that BMI and age were the strongest contributors to impurity reduction within the ensemble, with importance scores of 0.43 and 0.22, respectively. BI-RADS assessment followed as the next most influential predictor (0.16). These three variables collectively accounted for most of the model's decision structure.

Breast density, family history, biopsy history, and hormone therapy use provided smaller but non-trivial contributions, each with importance values between 0.01 and 0.06. Variables reflecting screening behavior, such as prior mammography and the presence of comparison films, had minimal importance, with values near zero, indicating that the Random Forest rarely used them to construct informative splits.

The ranking pattern reflects the algorithm's preference for continuous variables with wide numeric ranges, such as BMI and age, which naturally offer more opportunities for impurity reduction across tree splits. Figure 2 visualizes these importance values and highlights the dominance of continuous predictors within the ensemble. This contrasts with the coefficient-driven structure of Logistic Regression, where clinical history and radiologic assessment exerted stronger influence. The Random Forest model captured potential nonlinearities but did not translate this flexibility into improved discrimination, as its AUC remained below that of Logistic Regression.

Table 1
Logistic Regression Coefficient and Odds Ratio

Variable	Coefficient	Odds Ratio
biophx_c	0.582732	1.790924
famhx_c	0.368894	1.446135
mammtype	0.104228	1.109854
compfilm_c	0.097807	1.102750
hrt_c	0.057272	1.058944
bmi_c	0.028979	1.029403
age_c	0.026248	1.026596
density_c	-0.072932	0.929664
prvmam_c	-0.207761	0.812401
assess_c	-2.270356	0.103275

Table 2
Random Forest Feature Importance

Variable	Importance
bmi_c	0.434888
age_c	0.223706
assess_c	0.156083
density_c	0.062241
mammtype	0.031617
biophx_c	0.029553
famhx_c	0.027293
hrt_c	0.016681
compfilm_c	0.014637
prvmam_c	0.003301

Figure 1
Odds Ratios for Predictors in the Logistic Regression Model

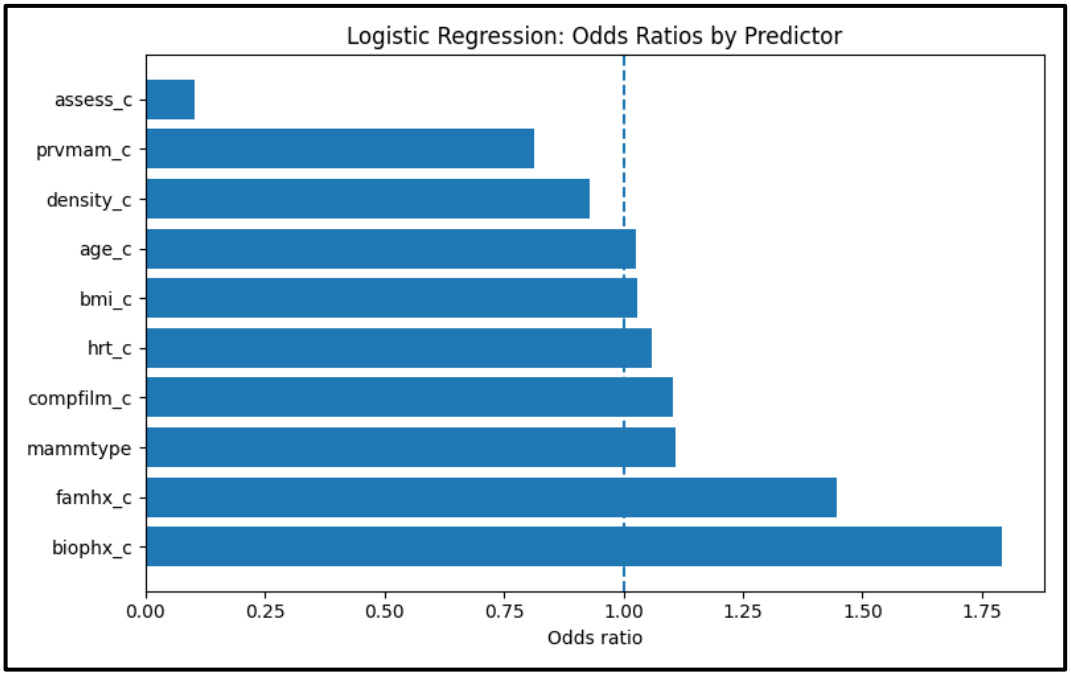
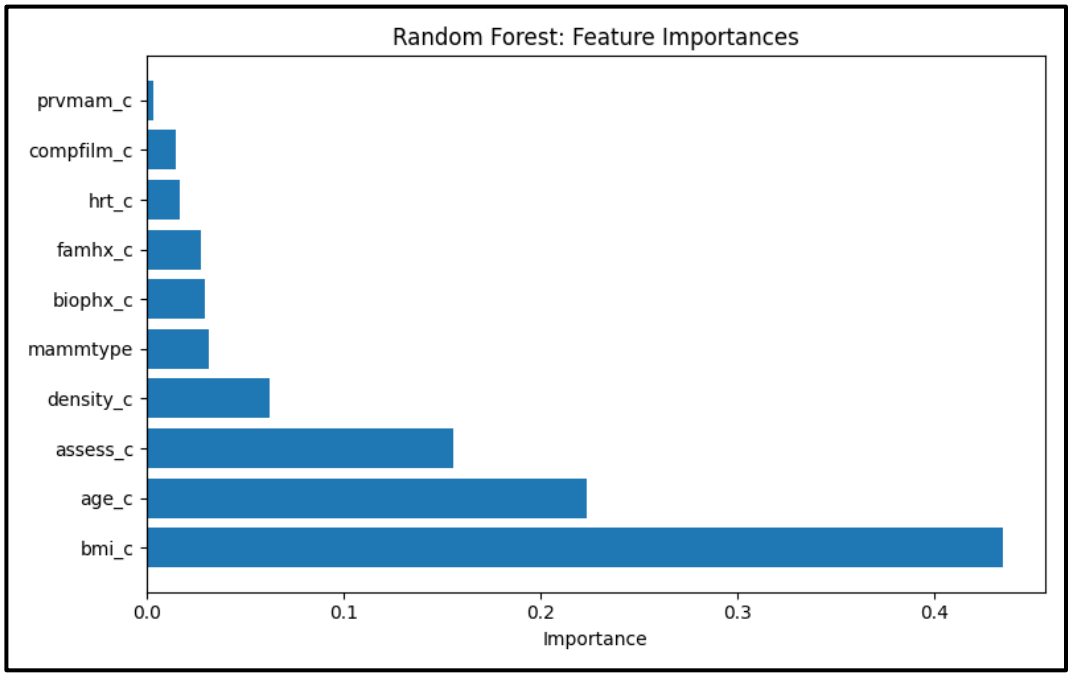


Figure 2
Random Forest Feature Importances for One-Year Breast Cancer Outcome



Discussion

Logistic Regression produced higher discrimination than Random Forest. The linear specification captured the dominant clinical relationships, with prior biopsy and family history exerting the strongest influence on the one-year outcome. The effect pattern is aligned with mammography literature in which biopsy history and family history consistently signal elevated short-term risk (American Cancer Society, 2022; National Cancer Institute, 2025).

Biopsy history exerted one of the strongest effects in the model because a prior biopsy often indicates previous abnormal findings or underlying tissue characteristics associated with elevated malignancy risk. Its coefficient produced a substantial increase in the log-odds of cancer, reflecting a well-documented clinical relationship. Family history also showed a similarly strong positive association. Individuals with one or more first-degree relatives diagnosed with breast cancer carry an increased baseline risk due to shared genetic susceptibility and familial patterns of disease. Logistic Regression captured these influences in a stable, interpretable manner, revealing that these two predictors were central in shaping short-term outcome probabilities. An interesting observation, BI-RADS behaved differently than expected from clinical diagnostic datasets. In this instructional screening dataset, BI-RADS showed a strong negative association because most cancers followed examinations coded as BI-RADS 0 (“needs additional imaging”), while higher BI-RADS categories (1–3) reflected routine benign screenings and were rarely associated with subsequent cancer diagnosis. As a result, increasing BI-RADS scores corresponded to a lower probability of one-year cancer within this dataset.

Random Forest performed slightly worse in discrimination but revealed a different structural pattern. The ensemble emphasized continuous variables, BMI and age, because tree-based partitioning favors predictors with wide numeric ranges and many potential split points. This produced a feature ranking that was different from the Logistic Regression model and illustrates how model architecture determines which variables appear most influential.

BMI emerged as a leading predictor in the forest because small changes across its broad numeric range creates many impurity-reducing partitions, even when its clinical association with short-term cancer risk is modest. Age followed a similar pattern. Its continuous structure allows repeated splitting, giving it disproportionate weight in the ensemble compared with its more limited effect in the linear model. These importance scores therefore reflect the mechanics of the algorithm rather than the strength of clinical associations. While Random Forest introduced flexibility to capture potential nonlinearities or interactions, the minimal improvement in accuracy suggests that such structure was limited in this dataset. The model functioned as a competent classifier but did not provide additional interpretive insight beyond what Logistic Regression already revealed.

Logistic Regression and Random Forest rank features differently because they operate on fundamentally different mathematical structures. The contrast in their rankings is informative and exposes how each model interprets the same dataset. Logistic Regression assigns importance based on linear contribution to the log-odds of the outcome, while Random Forest ranks features by how effectively they split the data to reduce impurity. Based on the results from these two models, it reinforces the idea that Supervised Learning methods are complementary tools for evaluating mammographic screening data. By applying models like Linear Regression and Random Forest to large observational datasets, this analysis demonstrates that using statistical learning can help in identifying consistent risk patterns as well as reduce reliance on subjective interpretation alone. The use of clinical, standardized datasets enables continued methodological research aimed at improving screening performance and reducing both misdiagnoses and overdiagnoses.

Limitations

The dataset used in this analysis was developed for instructional purposes rather than primary research, which does restrict the extent to which the results can be generalized beyond the context of modeled demonstrations. The available variables provide a structured representation of screening data

but do not capture the full clinical, technical, or imaging diversity present in real diagnostic environments. The dataset also contains no image files, limiting model development to radiologist-derived assessments and recorded patient characteristics rather than raw imaging information.

Data reduction during preprocessing further constrained the analysis. Although the original dataset contained approximately 40,000 mammography records, the removal of entries with missing or invalid values reduced the usable sample to roughly 15,000 examinations. This reduction narrowed the variability of several predictors and may have influenced model behavior, particularly for methods that depend on broad distributional structure. The analysis relied on a single train–test split without external validation, and performance results should be interpreted as internal estimates rather than indicators of broader predictive capacity.

Acknowledgements

“We thank the participating women, mammography facilities, and radiologists for the data they have provided. You can learn more about the BCSC at: http://www.bcsc-research.org/."

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Appendix

Table A1
BCSC's Digital Mammography Dataset Documentation

Variable Name	Description	Coding
age_c	patient's age in years at time of mammogram	Numerical
assess_c	Radiologist's assessment based on the BI-RADS scale	0 = Needs additional imaging 1 = Negative 2 = Benign finding(s) 3 = Probably benign 4 = Suspicious abnormality 5 = Highly suggestive of malignancy
cancer_c	binary indicator of cancer diagnosis within one year of screening mammogram	0 = no cancer diagnosis 1 = cancer diagnosis
compfilm_c	comparison mammogram from prior mammography examination available	0 = no 1 = yes 9 = missing
density_c	patient's BI-RADS breast density as recorded at time of mammogram	1 = Almost entirely fatty 2 = Scattered fibroglandular densities 3 = Heterogeneously dense 4 = Extremely dense
famhx_c	family history of breast cancer in a first degree relative	0 = no 1 = yes 9 = missing
hrt_c	current use of hormone therapy at time of mammogram	0 = no 1 = yes 9 = missing
prvmam_c	binary indicator of whether the woman had ever received a prior mammogram	0 = no 1 = yes 9 = missing
biophx_c	history of breast biopsy	0 = no 1 = yes 9 = missing
mammtype	film or digital mammogram	1 = film mammogram 2 = digital mammogram
CaTypeO	cancer type	1 = ductal carcinoma in situ 2 = invasive cancer 8 = no cancer diagnosis
bmi_c	body mass index at time of mammogram	Numerical or -99 if missing
ptid	patient's study id	